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EE101 - Basic Electronics

Pre-Tutorial Assignment 2: Solutions

1. Figure 1 shows an electric circuit with independent and dependent current sources. The numerical values of the circuit elements are as follows: $I_1 = 5A$, $I_2 = k \times i_1 = 5 \times i_1$, $R_1 = 1\Omega$, $R_2 = 2\Omega$, and $R_3 = 5\Omega$. Evaluate the branch currents i_1 , i_2 and i_3 using nodal analysis. The nodes **1**, **2**, and reference node **R** are marked in the figure.

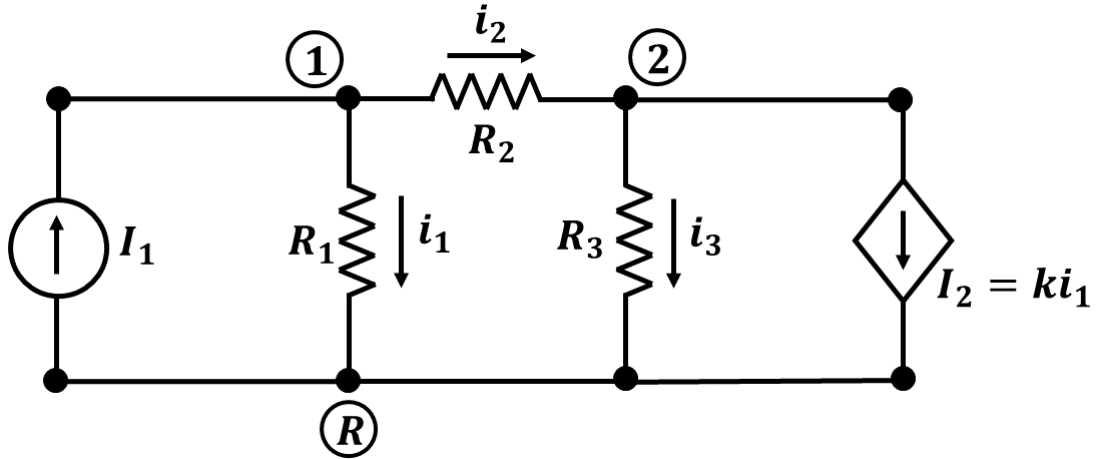


Figure 1: Electric circuit with independent and dependent current source.

Solution

Let v_1 and v_2 be the node voltages of the respective nodes **1** and **2** with respect to the reference node marked by **R**.

Application of KCL at Node **1** provides us with the following equation.

$$-\left(\frac{v_1 - v_2}{R_2}\right) - \frac{v_1}{R_1} + I_1 = 0 \quad (1)$$

Rearranging the terms, we obtain the following equation.

$$\left(\frac{1}{R_1} + \frac{1}{R_2}\right)v_1 - \left(\frac{1}{R_2}\right)v_2 = I_1 \quad (2)$$

Application of KCL at Node **2** provides us with the following equation.

$$\frac{v_1 - v_2}{R_2} - \frac{v_2}{R_3} - I_2 = 0 \quad (3)$$

We know that $I_2 = k \times i_1 = k \times v_1/R_1$. This substitution and after rearranging terms, we obtain the following equation.

$$\left(\frac{1}{R_2} - \frac{k}{R_1}\right)v_1 - \left(\frac{1}{R_2} + \frac{1}{R_3}\right)v_2 = 0 \quad (4)$$

After substitution of numerical values in Equations 2 and 4, we obtain the following equations.

$$3v_1 - v_2 = 10 \quad (5)$$

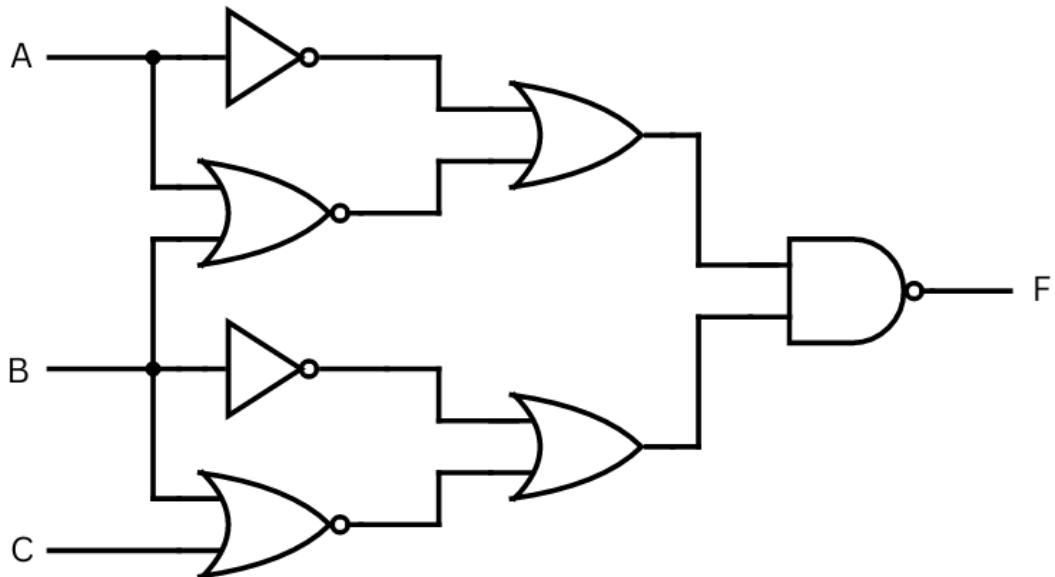
$$45v_1 + 7v_2 = 0 \quad (6)$$

Solving the above equations, we get $v_1 = 1.06V$ and $v_2 = -6.814V$.

The branch currents are thus obtained by Ohm's law.

$$\begin{aligned} i_1 &= \frac{v_1}{R_1} = \frac{1.06}{1}A = 1.06A \\ i_2 &= \frac{v_1 - v_2}{R_2} = \frac{1.06 + 6.814}{2}A = 3.937A \\ i_3 &= \frac{v_3}{R_3} = \frac{-6.814}{5}A = -1.363A \end{aligned} \quad (7)$$

2. Consider the logic circuit shown below in Figure 2.



(a) Express $F(A, B, C)$ in the Standard SOP form.

(b) Express $F(A, B, C)$ in the Standard POS form.

Solution

The truth table for the function $F(A, B, C)$ is as below:

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

(a) Standard SOP: $F(A, B, C) = \overline{A}.B.\overline{C} + \overline{A}.B.C + A.\overline{B}.\overline{C} + A.\overline{B}.C + A.B.\overline{C} + A.B.C$

(b) Standard POS: $F(A, B, C) = (A + B + C).(A + B + \overline{C})$